

Cloud Service Research

Objective: Understand cloud platforms and prepare service analysis.

WHAT IS CLOUD COMPUTING?

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, software, and analytics over the internet (“the cloud”) instead of local hardware. Instead of storing data in your computer, you store it on remote servers and access it anytime via internet.

Example

- Google Drive
- Dropbox
- Netflix

MAJOR CLOUD PROVIDERS

There are some major Cloud providers.

1. Amazon Web Services (AWS)

- Most widely used cloud platform
- Offers 200+ services
- Strong in scalability and enterprise solutions

Key Services:

- EC2 (Virtual Servers)
- S3 (Storage)
- RDS (Databases)

2. Microsoft Azure

- Strong integration with Windows and Microsoft tools
- Popular in businesses and government

Key Services:

- Azure Virtual Machines
- Azure Blob Storage
- Azure AI services

3. Google Cloud

- Best for AI, ML, and data analytics
- Used by companies like YouTube & Spotify

Key Services:

- Google Compute Engine
- BigQuery
- Cloud Storage

COMPARISON TABLE

Feature	AWS	Azure	Google Cloud
Market Share	Highest	Second	Third
Ease of use	Medium	Easy	Easy
AI/ML Strength	Good	Good	Best
Pricing	Paid	Hybrid	Competitive
Best for	Enterprises	Microsoft users	AI/Data projects

TYPES OF CLOUD SERVICES

There are given types of cloud service.

1.IaaS (Infrastructure as a Service)

- Virtual machines, storage, networks
- Example: AWS EC2

2.PaaS (Platform as a Service)

- Platforms to build apps
- Example: Google App Engine

3.SaaS (Software as a Service)

- Ready-to-use software
- Example: Gmail, Google Drive

ADVANTAGES OF CLOUD COMPUTING

- Low cost (no hardware needed)
- Scalable (increase/decrease resources)
- Remote access

- Data backup & recovery
- Security (advanced protection systems)

CHALLENGES / LIMITATIONS

- Internet dependency
- Privacy concerns
- Vendor lock-in
- Downtime risks

REAL-WORLD USE CASES

- Banking systems
- Netflix streaming
- Online education platforms
- AI training systems
- E-commerce (Amazon, Shopify)

CONCLUSION

Cloud computing is a powerful technology that allows users and organizations to access computing resources over the internet. AWS, Azure, and Google Cloud are the leading platforms offering scalable, secure, and efficient solutions for modern digital systems.